UNIT - I

Control System Modelling

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Control System

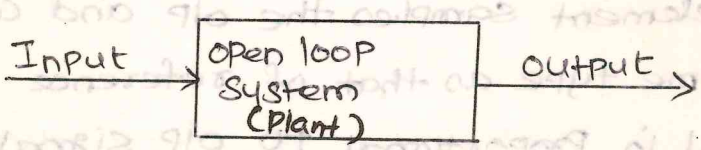
In a system when the output quantity is controlled by varying the input quantity, the system is called control system.

System

No. of components are connected in a sequence to perform a specific operation is called system.

Open Loop System

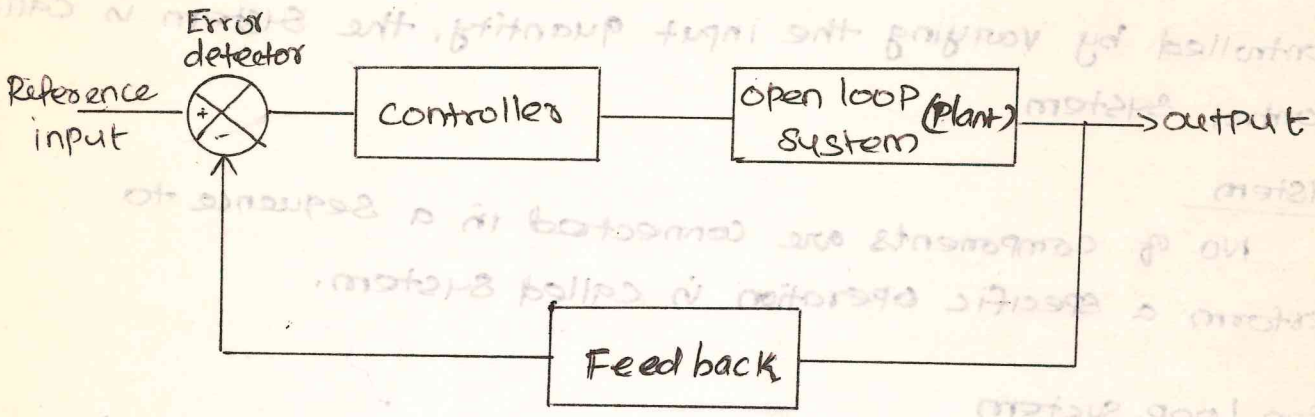
Any Physical system which does not automatically corrected the variation in its output, is called an open loop system, or control system in which the o/p quantity has no effect upon the input quantity are called open-loop control system. This means that the o/p is not feedback to the input for correction



In open loop system the o/p can be varied by varying the i/p. Due to external disturbances the o/p may be change. when the o/p changes due to disturbances, it is not followed by changes in input to correct the output. In open loop systems the changes in o/p are corrected by changing the i/p automatically.

Closed loop system

Control system in which the o/p has an effect upon the i/p quantity in such as manner has to maintain the desired o/p value are called closed loop system.



The open loop system can be modified as closed loop system by providing a feedback. The provision of feedback automatically corrects the the changes in o/p due to disturbances. The closed loop system is also called as automatic control system.

The reference signal corresponds to desired output. The feedback path element samples the o/p and converts into a signal of same type as that of reference signal. The feedback signal is proportional to o/p signal and it is fed to the error detector. The error signal generated by the error detector is the difference blw ref signal & feedback signal. The controller modifies & amplifies the error signal to produce better control action. It is fed to plant to correct its o/p.

Advantages

Open loop system

* Simple & economical.
* Easier to construct
* Generally system are stable

closed loop system

* System are accurate
* It is accurate even in the presence of non-linearities
* Systems are less affected by noise

Disadvantages

Open loop system

* Systems are inaccurate and unreliable
* due to external disturbances o/p will change & not corrected automatically.

closed loop system

* Systems are complex and costlier
* feed back in closed loop system may lead to oscillatory response
* feed back reduces the overall system gain
* More care is needed to design a stable closed loop system.

Examples of Control Systems

* Temperature control system
* Traffic control system
* Numerical control system
* Position control system using servomotor

Control System Terminology

Desired Response – the idealised instantaneous behaviour that we would like from the system.

Transient Response – the gradual change in the system as it approaches its approximation of the desired response.

Steady-State Response – the response of the system once it has finished changing and is now approximating the desired response.

Error – the difference between the input and the output of the system.

Steady-State Error – the difference between the steady-state response and the desired response.

Stability – the ability of the system to settle into a steady-state response.

Controller – the part of the system that generates the input to the plant or process being controlled.

Open-Loop – a system that does not monitor its output. Open-loop systems can not correct for disturbances.

Closed-Loop – a system that monitors its output and makes corrections to reduce error. By monitoring the output the system can correct for disturbances.

Disturbance – a signal that is not modelled or calibrated in the system leading to corruption of the expected behaviour.

Compensator – a system inserted into the controller to improve performance.

Feedback – a path that allows signals from the output of some sub-system to flow back and affect the input of some sub-system earlier in the system signal path.

Robust – a system that will still work as expected with changes to the system parameters, as might be caused by wear of components, or a change in behaviour with temperature.

Transfer function

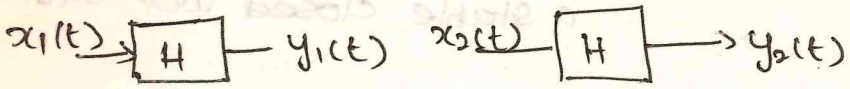
Transfer function of a system is designed as the ratio of laplace transform of o/p to laplace transform of i/p with zero initial conditions.

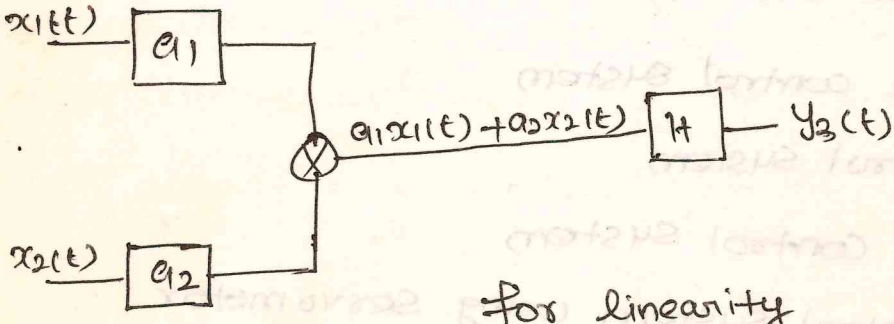
$$H(s) = \frac{Y(s)}{X(s)} \Big|_{\text{zero initial conditions.}}$$

Mathematical Models of Control System

The mathematical models of a control system constitutes a set of differential equations. The o/p of the system can be studied by solving the differential equations for various i/p conditions.

The mathematical model of a system is linear if it obeys the superposition principle.





for linearity

$$y_3(t) = a_1 x_1(t) + a_2 x_2(t)$$

A mathematical model will be linear if the differential equation describing the system has constant co-efficient, then the model is linear time invariant.

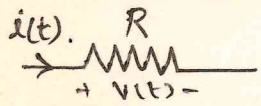
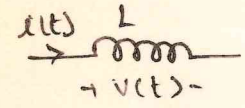
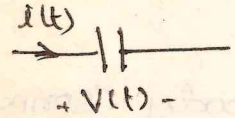
Note: * The differential equations of a linear time invariant [LTI] can be reshaped into different form for the convenience of analysis.

Ex: One such model for single i/p & single o/p system analysis is transfer function of the system.

Modelling of Electric system

The models of electrical systems can be obtained by using resistors, capacitor and inductor: For modelling electrical system the electrical network or equivalent circuit is formed by using R, L, & C and voltage or current source.

current - voltage relation

Element	Voltage across the element	current through the element
	$V(t) = R\, i(t)$	$i(t) = \frac{V(t)}{R}$
	$V(t) = L \frac{d\, i(t)}{dt}$	$i(t) = \frac{1}{L}\int V(t)\, dt$
	$V(t) = \frac{1}{C}\int i(t)\, dt$	$i(t) = C \frac{d}{dt} V(t)$

Mechanical Translational System

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Three basic Elements

* Mass
* spring
* dash-pot

Mass: Weight of the ~~Mechnical~~ Mechnical system is represented by mass.

Spring: The Elastic deformation of the body → It stores Energy during the Variation of its shape.

dashpot: The friction existing in mechnical system (Damper)
(Damping can be added to the system by using dashpot)

Newton 2nd law

Sum of applied force is equal to the sum of opposing force on a body.

So, when a force applied to a translational system it is opposed by opposing force due to Mass, friction and elasticity of the system.

$x \rightarrow$ Displacement, m

$V \rightarrow \frac{dx}{dt}$ = velocity, m/sec

$a \rightarrow \frac{dv}{dt} = \frac{d^2x}{dt^2}$ = Acceleration m/sec^2

$f \rightarrow$ Applied force N

$f_m \rightarrow$ opposing force offered by mass of the body

$f_k \rightarrow$ " " " " ~~spring~~ elasticity of the body (spring)

$f_b \rightarrow$ " " " " friction of the body (dashpot)

$M \rightarrow$ Mass (kg)

$K \rightarrow$ Stiffness of spring N/m

$B \rightarrow$ Viscous friction co-efficient N-sec/m

Translational sys → Force balanced equation

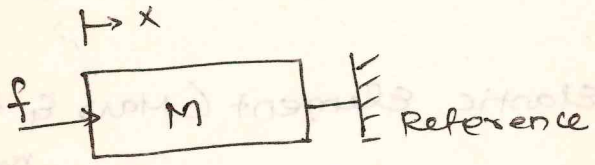
Rotational system → Torque balanced equation

Force balanced equation of idealized element

Mass

Consider ideal mass element (friction & elasticity negligible)

Force be applied on it, mass will offer an opposing force which is proportional to acceleration of the body.



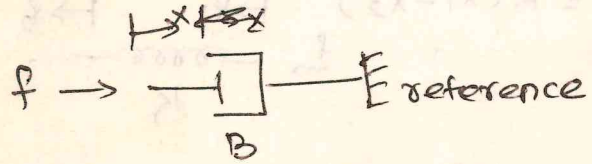
$f_m \propto \frac{d^2x}{dt^2}$ (or) $f_m = M \cdot \frac{d^2x}{dt^2}$

Newton's second law → $f = f_m = M \cdot \frac{d^2x}{dt^2}$

Dash pot

Consider a ideal dash-pot (or) friction Element (mass & spring negligible)

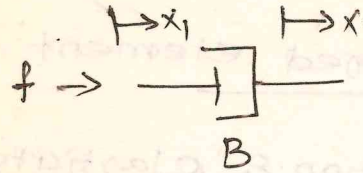
Force be applied on it, dash-pot will offer an opposing force which is proportional to velocity of the body



$f_b \propto \frac{dx}{dt}$ (or) $f_b = B \cdot \frac{dx}{dt}$

Newtons second law $f = f_b = B\frac{dx}{dt}$

when the dash-pot has displacement at both ends the opposing force is proportional to differential velocity



$$f_b \propto \frac{d}{dt}(x_1 - x_2) \quad \text{i.e} \quad f_b = B\frac{d}{dt}(x_1 - x_2)$$

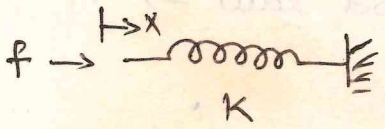
$$f = f_b = B\frac{d}{dt}(x_1 - x_2)$$

Spring

Consider an ideal Elastic element (Mass & friction negligible)

Force be applied on it, spring will offer opposing force which is proportional to displacement of the body

$f_K \propto x \quad : \quad f_K = Kx$

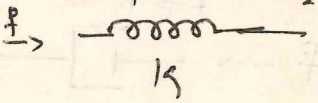


Newton's second law $f = f_K = Kx$

when the spring has displacement at both ends, the opposing force is proportional to differential displacement

$f_K \propto (x_1 - x_2) \quad ; \quad f_K = K(x_1 - x_2)$

$\therefore f = f_K = K(x_1 - x_2)$



Guidelines to determine the transfer function of mechanical translational system

* The differential equation governing the system are obtained by writing force balance equation at nodes in the system. The nodes are meeting point of elements. Generally the nodes are mass element in the system. In some cases the nodes may be without mass Element.

* The linear displacement of nodes (masses) are assumed as x_1, x_2, x_3 etc, and assign a displacement to each node (mass). The first derivative of the displacement is velocity & the second derivative of the displacement is accelearation.

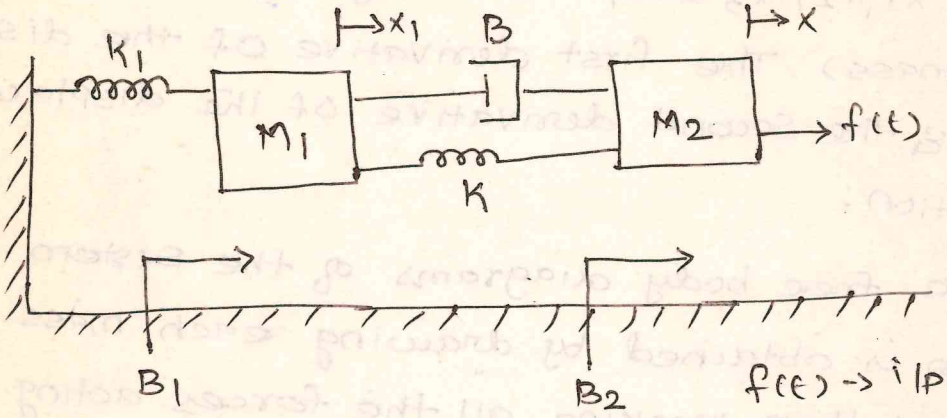
* Draw the free body diagrams of the system. The free body diagram is obtained by drawing each mass separately and then marking all the forces acting on that mass ~~(node)~~(node). Always the opposing force acts in a direction opposite to applied force. The mass has to move in the direction of the applied force. Hence the displacement, velocity and accelearation of the mass will be in the direction of the applied force. If there is no applied force then the displacement, velocity and accelearation of the mass is in a direction opposite to that of opposing force.

* For each free body diagram write one differential equation by equating the sum of applied forces to the sum of opposing forces.

* Take laplace transform of differential equations to convert them to algebric equation. Then rearrange the S-domain equation to eliminate the unwanted variables and obtain the ratio blw o/p variable to i/p variable. This ratio is the transfer function of the system.

Problems

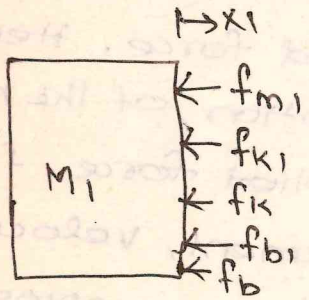
1) write the differential equation governing the given mechanical system & determine the transfer function.



f(t) -> i/p applied force

$T.F = \frac{X(s)}{F(s)}$ X - displacement o/p

Free body diagram of Mass M_1



opposing force acting on a mass is marked as $f_{m_1}, f_{b_1}, f_b, f_{k_1}$ & f_k

$$f_{m_1} = M_1 \frac{d^2x_1}{dt^2} \quad ; \quad f_{b_1} = B_1 \frac{dx_1}{dt} \quad ; \quad f_{k_1} = K_1 x_1$$

$$f_b = B \frac{d}{dt}(x_1 - x) \quad ; \quad f_k = K(x_1 - x)$$

By Newton's law

$$f_{m_1} + f_{B_1} + f_b + f_{K_1} + f_K = 0$$

$$M_1 \frac{d^2x_1}{dt^2} + B_1 \frac{d}{dt} x_1 + K_1 x_1 + B \cdot \frac{d}{dt}(x_1 - x) + K(x_1 - x) = 0$$

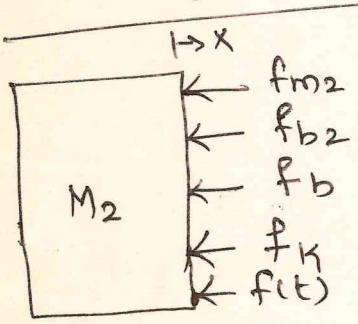
on taking LT

$$M_1 s^2 X_1(s) + B_1 s \cdot X_1(s) + K_1 X_1(s) + Bs[X_1(s) - X(s)] + K[X_1(s) - X(s)] = 0$$

$$X_1(s)\left[M_1 s^2 + [B_1 + B]s + (K_1 + K)\right] - X(s)[Bs + K] = 0$$

$$X_1(s) = \frac{X(s) \cdot (Bs + K)}{M_1 s^2 + (B_1 + B)s + (K_1 + K)} \quad \text{— (1)}$$

Free body diagram of mass M_2



$$f_{m_2} = M_2 \frac{d^2x_2}{dt^2} \; ; \; f_{b_2} = B_2 \frac{dx_2}{dt} \; ; \; f_K = K(x_2 - x_1)$$

$$f_b = B\frac{d}{dt}(x_2 - x_1)$$

Newtons law

$$f_{m_2} + f_{b_2} + f_b + f_K = f(t)$$

$$M_2 \frac{d^2x_2}{dt^2} + B_2 \frac{dx_2}{dt} + B(x_2 - x_1) + K(x_2 - x_1) = f(t)$$

on taking LT

$$M_2 s^2 X_2(s) + B_2 s \cdot X_2(s) + Bs[X_2(s) - X_1(s)] + K[X_2(s) - X_1(s)] = F(s)$$

$$X_2(s)\left[M_2 s^2 + (B_2 + B)s + K\right] - X_1(s)[Bs + K] = F(s) \quad \text{— (2)}$$

Sub eqn (1) i.e $X_1(s)$ in the eqn (2)

$$X(s)\left[M_2 s^2 + (B_2 + B)s + K\right] - \frac{X(s) \cdot (Bs + K)^2}{M_1 s^2 + (B_1 + B)s + (K_1 + K)} = F(s).$$

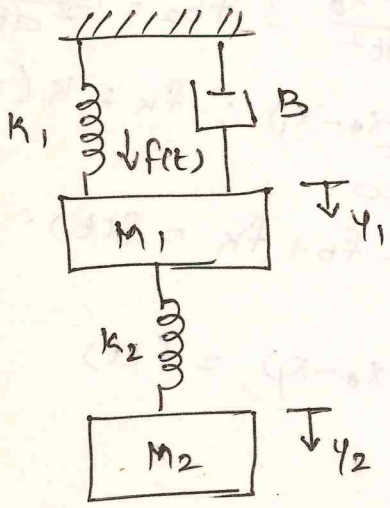
$$X(S)\left[\frac{M_1S^2+(B_1+B)S+(K_1+K)\cdot[M_2S^2+(B_2+B)S+K]-(BS+K)^2}{M_1S^2+(B_1+B)S+(K_1+K)}\right]:F(S)$$

T.F

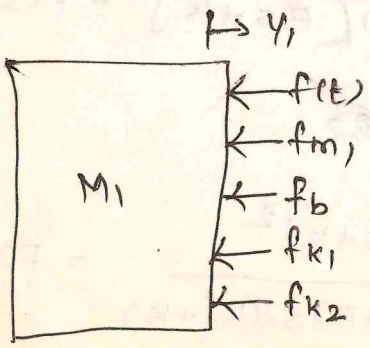
$$\frac{X(S)}{F(S)}=\frac{M_1S^2+(B_1+B)S+(K_1+K)}{[M_1S^2+(B_1+B)S+(K_1+K)][M_2S^2+(B_2+B)S+K)]-(BS+K)^2}$$

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2) Determine the T.F $\frac{Y_2(S)}{F(S)}$ of the given system



Free body diagram of Mass 1



$f_{m_1}=M\cdot\frac{d^2y_1}{dt^2}$; $f_b=B\frac{d}{dt}y_1$; $f_{k_1}=K_1y_1$

$f_{k_2}=K_2(y_1-y_2)$

Newtons law

$f_{m_1}+f_b+f_{k_1}+f_{k_2}=f(t)$

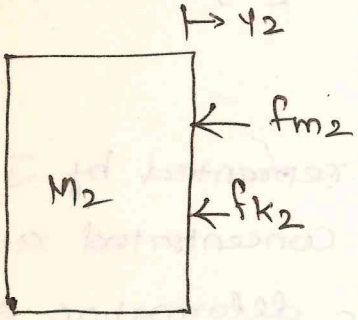
$$M_1 \frac{d}{dt^2} y_1 + B \frac{d}{dt} y_1 + K_1 y_1 + K_2 (y_1 - y_2) = f(t)$$

on taking LT

$$M_1 S^2 Y_1(s) + B \cdot S \cdot Y_1(s) + K_1 Y_1(s) + K_2 [Y_1(s) - Y_2(s)] = F(s)$$

$$Y_1(s) [M_1 S^2 + BS + (K_1 + K_2)] - Y_2(s) K_2 = F(s) \quad -①$$

Free body diagram of Mass M_2



$f_{m_2} = M_2 \frac{d^2}{dt^2} y_2$; $f_{k_2} = K_2 \cdot (y_2 - y_1)$

Newtons second law

$f_{m_2} + f_{k_2} = 0$

$$M_2 \frac{d}{dt} y_2 + K_2 (y_2 - y_1) = 0$$

on taking LT

$$M_2 S^2 Y_2(s) + K_2 [Y_2(s) + Y_1(s)] = 0$$

$$Y_2(s) [M_2 S^2 + K_2] - K_2 Y_1(s) = 0$$

$$Y_1(s) = \frac{Y_2(s) \cdot [M_2 S^2 + K_2)}{K_2} \quad -②$$

Sub eqn ② in eqn ①

$$Y_2(s) \left[\frac{M_2 S^2 + K_2}{K_2} \right] \left[M_1 S^2 + BS + (K_1 + K) \right] - K_2 Y_1(s) = F(s)$$

$$Y_2(s) \left[\frac{(M_2 S^2 + K_2)(M_1 S^2 + BS + (K_1 + K)) - K_2^2}{K_2} \right] = F(s)$$

$$\frac{Y_2(s)}{R(s)} = \frac{K_2}{\left[M_1 s^2 + Bs + (K_1 + K_2)\right]\left[M_2 s^2 + K_2\right] - K_2^2}$$

Mechanical rotational System

1) Moment of inertia of mass [J]
2) Dash pot with frictional co-efficient [B]
3) torsional spring with stiffness [K]

The weight of the rotational system is represented by J. This J of the system is considered to be concentrated at the center of gravity of the body. The elastic deformation of the body represented by a spring & friction existing in rotational system represented by dash-pot. The dash-pot is the piston rotating inside a cylinder filled with viscous fluid.

When a torque is applied to a rotational system, it is opposed by opposing torques due to moment of inertia, friction and elasticity of the body. The torque acting on a rotational system. is governed by the Newton's second law.
It is states the sum of torques acting on a body is zero (ie. sum of applied torque is equal to sum of opposing torque on a body)

List of symbols used in Mech. rotational systems

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θ - Angular displacement, rad

$\frac{d\theta}{dt}$ → Angular velocity rad/sec

$\frac{d^2\theta}{dt^2}$ → Angular acceleration rad/sec^2

T → Applied torque

J → moment of inertia Kg-m^2/rad

B → Rotational frictional co-efficient N-m/rad/sec

K → Stiffness of the spring N-m/rad.

Torque balance equation of ideal elements

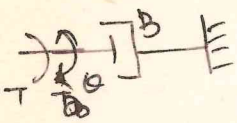
Moment of Inertia.

T_j → opposing torque due to moment of inertia $\propto$ angular acceleration.

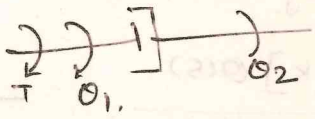


$T_j \propto \frac{d^2\theta}{dt^2}$ $\quad T_j = J\frac{d^2\theta}{dt^2}$.

Ideal frictional element Dashpot. $\propto$ angular velocity of the body.

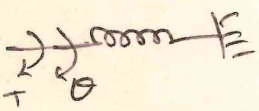


$T_b \propto \frac{d\theta}{dt}$ $\quad \therefore T_b = B\frac{d\theta}{dt}$.

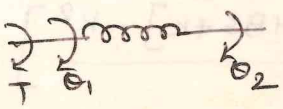


$T_b \propto \frac{d}{dt}(\theta_1 - \theta_2)$ $\quad \therefore T_b = B\cdot\frac{d}{dt}(\theta_1 - \theta_2)$

Ideal spring. $\propto$ angular displacement of the body.

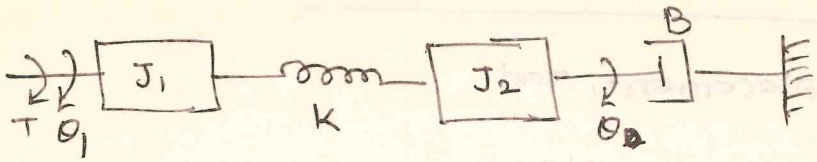


$T_k \propto \theta$ $\quad \therefore T = K\theta$.

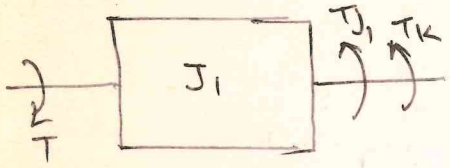


$T_k \propto (\theta_1 - \theta_2)$ $\quad \therefore T = K(\theta_1 - \theta_2)$

1. Obtain differential equation and transfer function of the sys. (15)



Transfer function $\frac{\theta(s)}{T(s)}$.



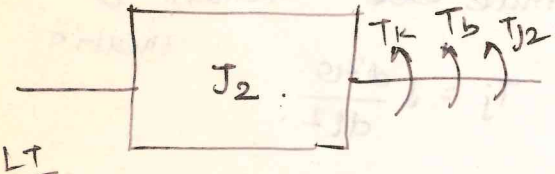
$$T_{J_1} + T_K = T$$

$$J_1 \frac{d^2\theta_1}{dt^2} + K(\theta_1 - \theta) = T$$

LT

$$J_1 s^2 \theta_1(s) + K\theta_1(s) - K\theta(s) = T(s).$$

$$\theta_1(s)\left[J_1 s^2 + K\right] - K\theta(s) = T(s) \quad -(1)$$



$$T_K + T_b + T_{J_2} = 0$$

$$J_2 \frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt} + K(\theta - \theta_1) = 0$$

LT

$$J_2 s^2\theta(s) + B \cdot s \cdot \theta(s) + K\theta(s) - K\theta_1(s) = 0$$

$$\theta(s)\left[J_2 s^2 + Bs + K\right] - K\theta_1(s) = 0$$

$$\theta_1(s) = \frac{\left[J_2 s^2 + Bs + K\right]\theta(s)}{K} \quad -(2)$$

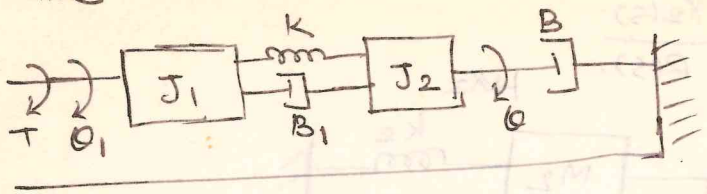
Sub eqn (2) in (1)

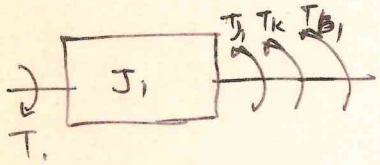
$$\left[J_1 s^2 + K\right]\left[\frac{J_2 s^2 + Bs + K}{K}\theta(s)\right] - K\theta(s) = T(s).$$

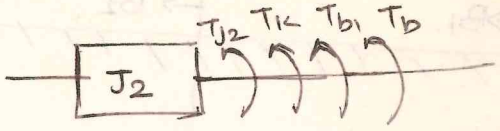
$$\theta(s)\left[\frac{\left[J_1 s^2 + K\right]\left[J_2 s^2 + Bs + K\right] - K^2}{K}\right] = T(s).$$

$$\boxed{\frac{\theta(s)}{T(s)} = \frac{K}{\left[J_1 s^2 + K\right]\left[J_2 s^2 + Bs + K\right] - K^2}}$$

2) Determine the transfer function.







$T_{J_1} + T_k + T_{b_1} = T$

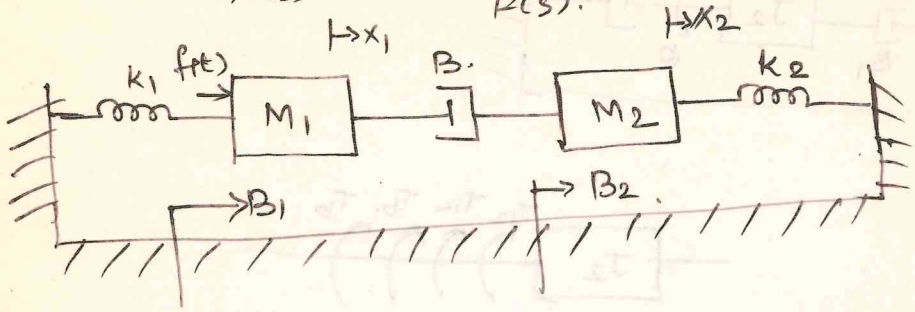
$T_{J_2} + T_k + T_{b_1} + T_b = 0$

$\Theta_1(s)\left[J_1 s^2 + B_1 s + k\right] - \Theta(s)\left[B_1 s + k\right] = T(s).$

$\Theta_1(s) = \left[\dfrac{J_2 s^2 + (B_1 + B)s + k}{B_1 s + k}\right]\Theta(s)$

$$\frac{\Theta(s)}{T(s)} = \frac{B_1 s + k}{\left[J_1 s^2 + B_1 s + k\right]\left[J_2 s^2 + (B_1 + B)s + k\right] - (B_1 s + k)^2}$$

1) Obtain TF. $\frac{X_1(s)}{F(s)}$ and $\frac{X_2(s)}{F(s)}$.

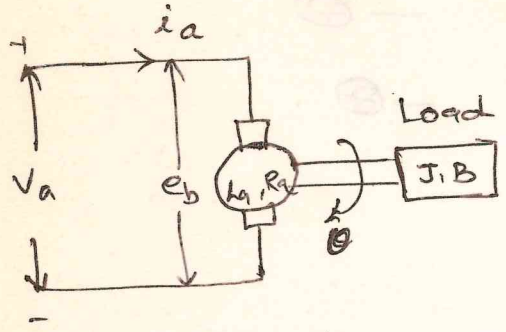


$$\frac{X_1(s)}{F(s)} = \frac{M_2 s^2 + (B_2 + B)s + K_2}{[M_1 s^2 + (B_1 + B)s + K_1][M_2 s^2 + (B_2 + B)s + K_2] - (Bs)^2}$$

$$\frac{X_2(s)}{F(s)} = \frac{Bs}{[M_2 s^2 + (B_2 + B)s + K_2][M_1 s^2 + (B_1 + B)s + K_1] - (Bs)^2}$$

Transfer function of Armature Controlled DC motor

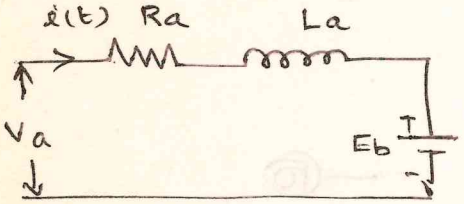
The mechanical system consist of the rotating part of the motor & load connected to the shaft of the motor



K_t - Torque constant N-m/A
T - Torque N-m
θ - angular displacement of shaft, rad
J - moment of inertia
B - frictional co-efficient
K_b - Back emf constant

Equivalent ckt of armature

[voltage across the element]



by KVL

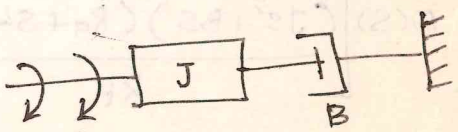
$$i_a R_a + L_a \frac{di_a}{dt} + e_b = V_a \quad -(1)$$

Torque of the DC motor directly proportional to flux & current

$$\therefore T \propto i_a \Rightarrow T = K_t i_a \quad -(2)$$

Equivalent ckt of mechanical system

$$J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt} = T \quad -(3)$$



Back emf of DC motor is proportional to speed of the motor [speed = angular velocity]

$$e_b \propto \frac{d\theta}{dt} \quad \therefore e_b = K_b \frac{d\theta}{dt} \quad -(4)$$

There are 4 differential equations governing the armature controlled DC motor speed control system

Taking LT

$$I_a(s) R_a + L_a \cdot s \cdot I_a(s) + E_b(s) = V_a(s) \quad -(5)$$

$$T(s) = K_t I_a(s) \quad -(6)$$

$$J s^2 \theta(s) + B \cdot s\, \theta(s) = T(s) \quad -(7)$$

$$E_b(s) = K_b \cdot s \cdot \theta(s) \quad -(8)$$

on equating eqn (6) & (7)

$$K_t I_a(s) = J s^2 \theta(s) + B s \theta(s)$$

$$I_a(s) = \frac{\theta(s)[J s^2 + B s]}{K_t} \quad -(9)$$

eqn (5) can be written as

$$I_a(s)\,[R_a + L_a \cdot s] + E_b(s) = V_a(s) \quad -(10)$$

Sub $I_a(s)$ & $E_b(s)$ value in eqn (5)

$$\frac{\theta(s)[J s^2 + B s]}{K_t}[R_a + s L_a] + K_b \cdot s \cdot \theta(s) = V_a(s)$$

$$\theta(s)\left[\frac{(J s^2 + B s)(R_a + s L_a) + K_b \cdot K_t \cdot s}{K_t}\right] = V_a(s)$$

$$\frac{\theta(s)}{V_a(s)} = \frac{K_t}{[(J s^2 + B s)(R_a + s L_a) + K_b K_t s]}$$

$$= \frac{K_t}{R_a\left[1 + \frac{s L_a}{R_a}\right] B s\left[1 + \frac{J s^2}{B s}\right] + K_b K_t s}$$

$$\frac{\theta(s)}{V_a(s)} = \frac{K_t / R_a B}{s\left[\left[1 + s T_a\right](1 + s T_m) + \frac{K_b K_t}{R_a B}\right]}$$

÷ by $R_a B$ both on & Nu.

$\frac{L_a}{R_a} = T_a$ – Electric time constant

$J/B = T_m$ = Mech Time constant

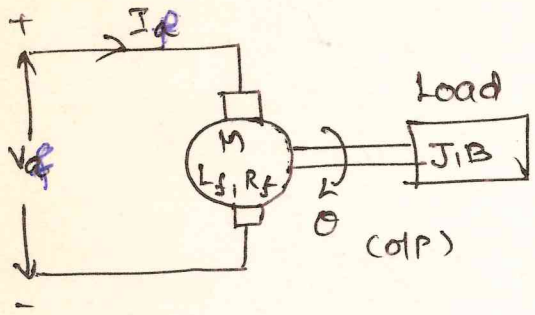
$$= \frac{K_t}{R_a J s^2 + R_a B s + L_a J s^3 + L_a B s^2 + K_b K_t s}$$

$$\frac{\theta(s)}{V_a(s)} = \frac{K_t}{s\left[J L_a s^2 + (J R_a + B L_a)s + B R_a + K_b K_t\right]}$$

Transfer function of Field control Dc shunt motor

---

flux is directionally proportional to field current.



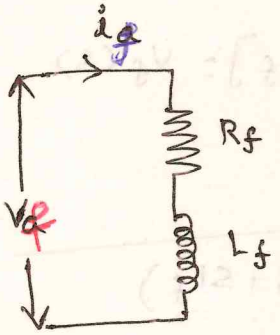
K_{tf} → Torque constant

J →

B →

KVL

$$R_f i_f + L_f \frac{di_f}{dt} = V_f \quad -①$$

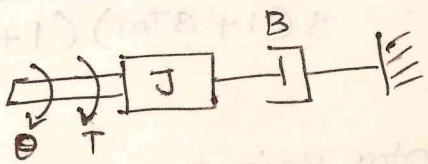


Torque is directly proportional to flux

$$T \propto i_f \; ; \; K = K_{tf} i_f \quad -②$$

Mech sys

$$J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt} = T \quad -③$$



on taking LT

$$R_f I_f(s) + L_f \cdot s\, I_f(s) = V_f(s) \quad -④$$

$$T(s) = K_{tf}\, I_f(s) \quad -⑤$$

$$Js^2\theta(s) + B \cdot s\,\theta(s) = T(s) \quad -⑥$$

equating eqn ⑤ & ⑥

$$K_{tf} I_f(s) = Js^2\theta(s) + B \cdot s\,\theta(s)$$

$$I_f(s) = \frac{[Js^2 + Bs]}{K_{tf}}\theta(s) \quad -⑦$$

eqn ④

$$I_f(s)\left[R_f + sL_f\right] = V_f(s) \quad -⑧$$

Sub equ ⑦ in equ ⑧

$$\frac{Js^2+Bs}{K_{tf}}\theta(s)\left[R_f + sL_f\right] = V_f(s)$$

$$\frac{\theta(s)}{V_f(s)} = \frac{K_{tf}}{(Js^2+Bs)(R_f+sL_f)}$$

$$= \frac{K_{tf}}{s(Js+B)R_f\left(1+\frac{sL_f}{R_f}\right)} \Rightarrow \frac{K_{tf}}{R_f s B\left(1+\frac{Js}{B}\right)\left(1+\frac{sL_f}{R_f}\right)}$$

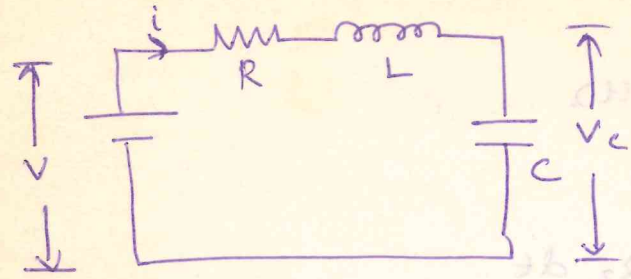
$$= \frac{K_{tf}\,|\,R_f B}{s(1+sT_m)(1+sT_f)} = \frac{K_m}{s(1+sT_m)(1+sT_f)}$$

K_m – motor gain constant $= K_{tf}\,|\,R_f B$

T_f – field time constant $= L_f\,|\,R_f$

T_m – Mech time constant $= J\,|\,B$

(1)



$V = V_R + V_L + V_C$

$= iR + L\frac{di}{dt} + \frac{1}{C}\int i\,dt$

$V(S) = RI(S) + LS\cdot I(S) + \frac{1}{CS} I(S)$

$V(S) = I(S)\left[R + LS + \frac{1}{CS}\right]$

$I(S) = \frac{V(S)}{R + LS + \frac{1}{CS}}$

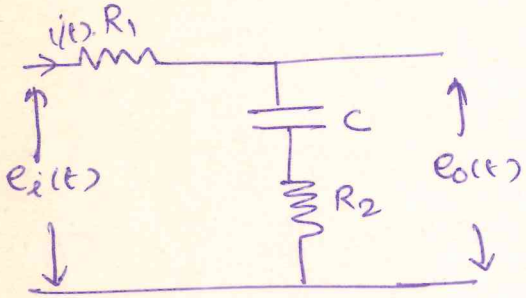
$V_C(S) = \frac{1}{CS} I(S)$

$= \frac{1}{CS}\left[\frac{V(S)}{R + LS + \frac{1}{CS}}\right]$

$$\frac{V_C(S)}{V(S)} = \frac{1}{LCS^2 + RCS + 1}$$

$$\frac{\frac{1}{CS}}{RCS + LCS^2 + 1}$$

(2)



$e_i(t) = R_1 i(t) + R_2 i(t) + \frac{1}{C}\int i(t)\cdot dt$

$e_o(t) = \frac{1}{C}\int i(t)\,dt + R_2 i(t)$

LT

$E_i(S) = R_1 I(S) + R_2 I(S) + \frac{1}{CS} I(S)$ — (1)

$E_o(S) = \frac{1}{CS} I(S) + R_2 I(S)$ — (2)

From eqn (1)

$I(S) = \frac{E_i(S)}{R_1 + R_2 + \frac{1}{CS}} \Rightarrow \frac{E_i(S)\,CS}{R_1 CS + R_2 CS + 1}$

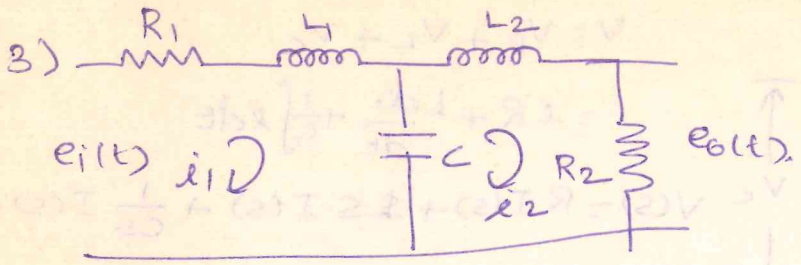
From eqn (2)

$E_o(S) = I(S)\left[R_2 + \frac{1}{CS}\right]$

$= I(S)\left[\frac{R_2 CS + 1}{CS}\right]$

$$\frac{E_o(S)}{E_i(S)} = \frac{R_2 CS + 1}{(R_1 + R_2)CS + 1}$$

3)



$$e_i(t) = R_1 i_1 + L_1 \frac{d}{dt} i_1 + \frac{1}{C} \int i_1 - i_2 \, dt$$

$$e_o(t) = L_2 \frac{di_1}{dt} + R_2 i_1 + \frac{1}{C} \int i_2 - i_1 \, dt$$

$$\frac{E_o(s)}{E_i(s)} = \frac{R_2 Cs}{[L_1 Cs^2 + R_1 Cs + 1][L_2 Cs^2 + R_2 Cs + 1] - 1}$$

SIGNAL FLOW GRAPH

The signal flow graph is used to represent the control system graphically & it was developed S.J. Mason. (1953).

Block diagrams are used for representation of the inter-relationship of the variable of a control system. But it is difficult to reduce the block diagram (i.e reduction procedure). An alternative method for determining the relationship blw system variable, called signal flow graph. It is based on representation of the system by line segment. The advantage of this method is availability of a gain rule.

Mason's Gain formula.

Overall gain $T = \frac{1}{\Delta} \sum_{k} P_k \Delta_k$.

P_k → Forward path gain of kth forward path

Δ_k → NO. of forward paths.

Δ → 1 − (sum of individual loop gains) + [sum of gain products of all possible combinations of two non-touching loops]

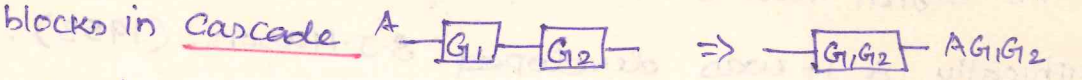
− [Sum of gain products of all possible combinations of three non-touching loops] + ---

Δ_k := Δ for that part of graph which is not touching k^{th} forward path.

Block diagram reduction.

Rules.

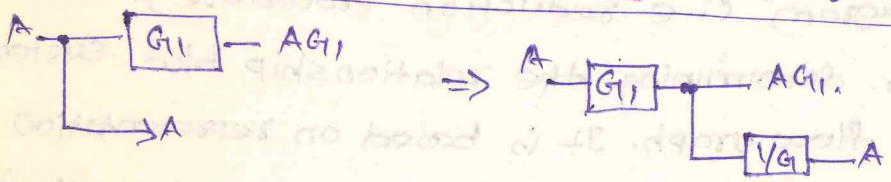
Combining the blocks in Cascade



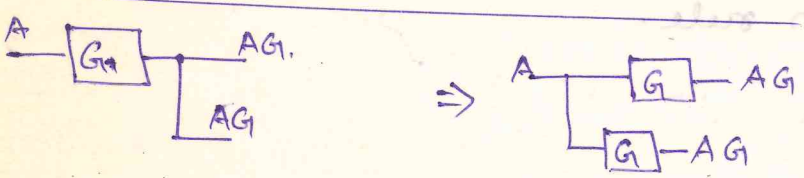
Combining the blocks in ||el



Moving the branch point in front (ahead) of the block.

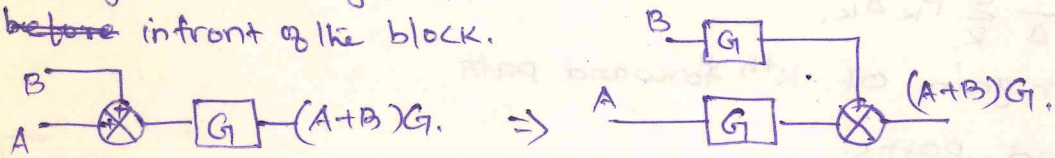


Moving the branch point before the block.

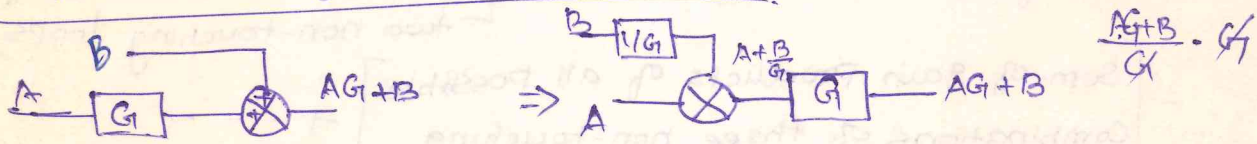


Moving the Summing point

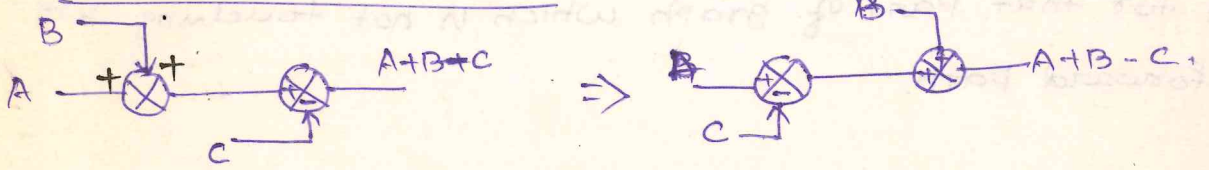
~~before~~ infront of the block.



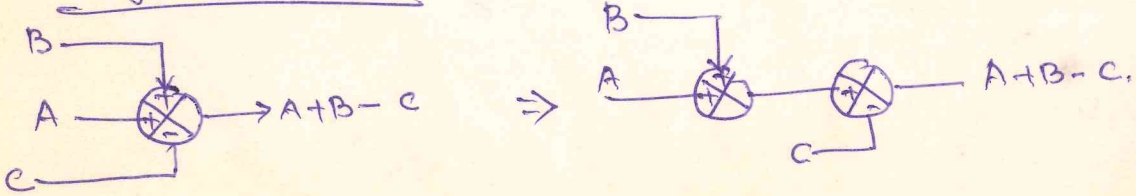
Moving the Summing point before the block.



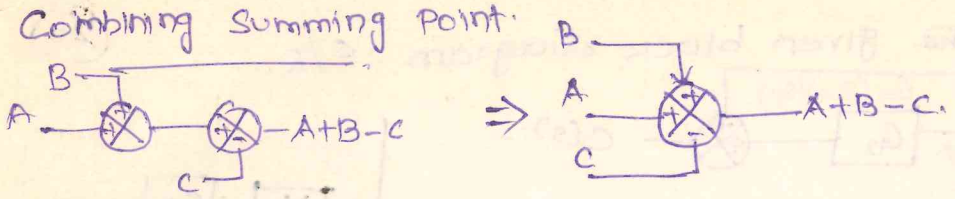
Interchanging Summing point.



Splitting Summing point.



Combining Summing Point:



Elimination of feedback loop. (negative)



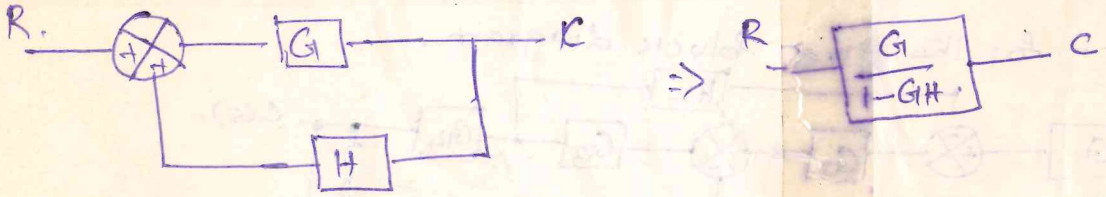
$\frac{C}{R}$: $C = (R - CH)G$

$C = RG - CHG$

$C(1+GH) = RG.$

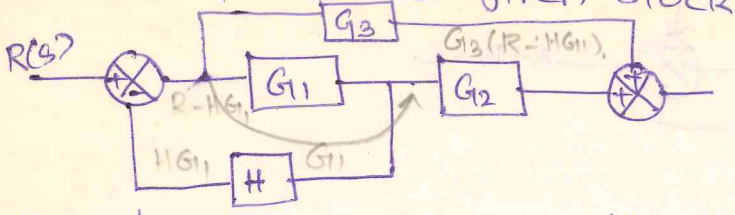
$\frac{C}{R} = \frac{G}{1+GH}.$

Elimination of positive feedback loop

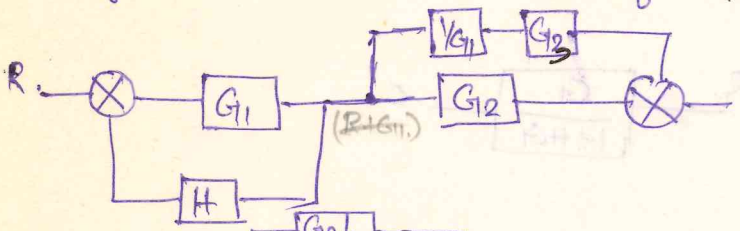


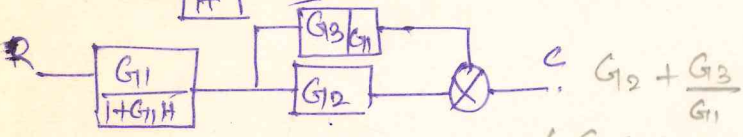
Dr. M. SUNDAR RAJAN, Ph.D.,
Associate Professor,
Industrial Control Stream,
Faculty of Elec. & Comp. Engg
AMIT, AMU.

1) Find the T.F for the given block diagram C/R.

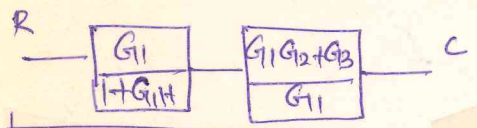


moving the branch point infront of block G_1.



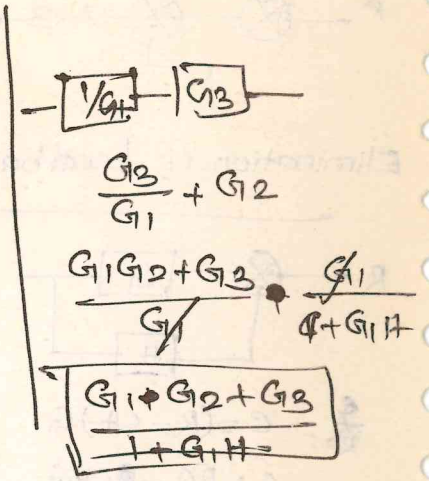


$G_2 + \frac{G_3}{G_1}$

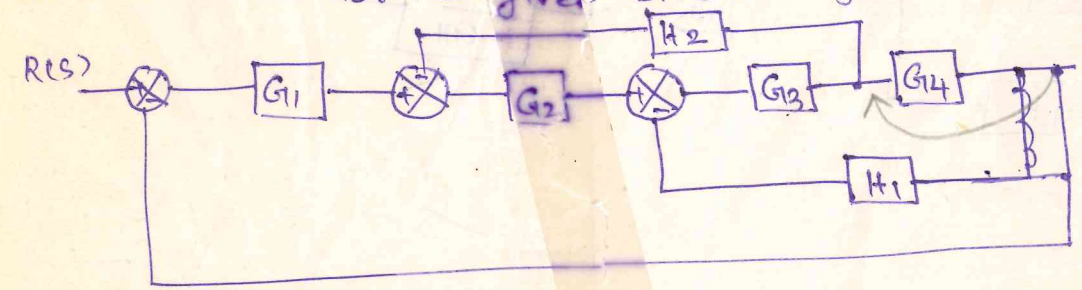


$$\left(\frac{G_1G_2+G_3}{G_1}\right)\left(\frac{G_1}{G_1H+1}\right) = \frac{G_1G_2+G_3}{1+G_1H}$$

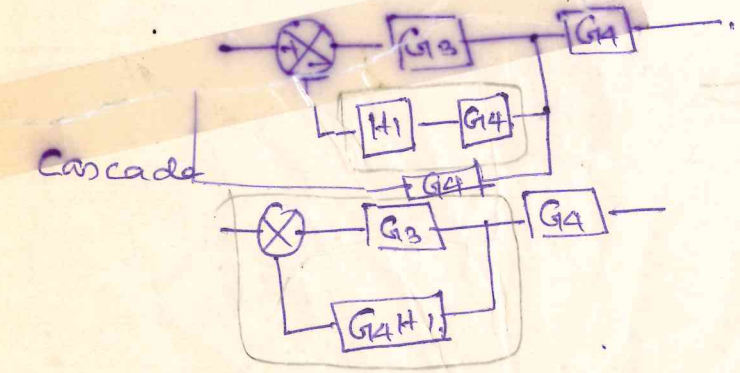
$$\frac{C}{R} = \frac{G_1G_2+G_3}{1+G_1H}$$



2). Find the T.F for the given block diagram.

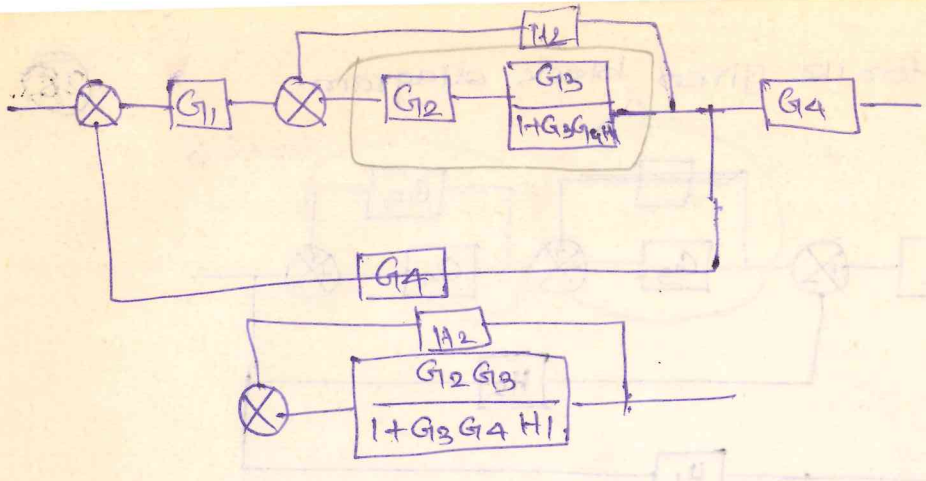


moving the b.p before the block.



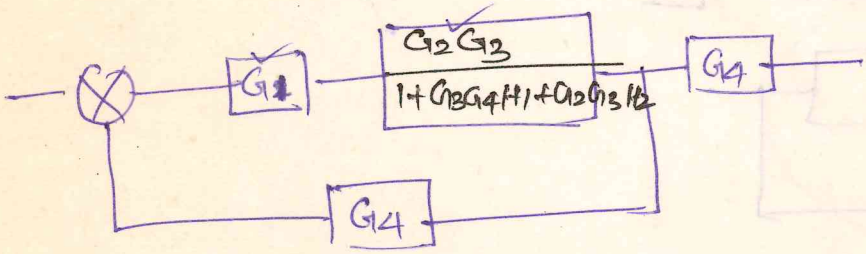
Eliminating feedback path.





$$G_2 \cdot \frac{G_3}{1+G_3G_4H_1}$$

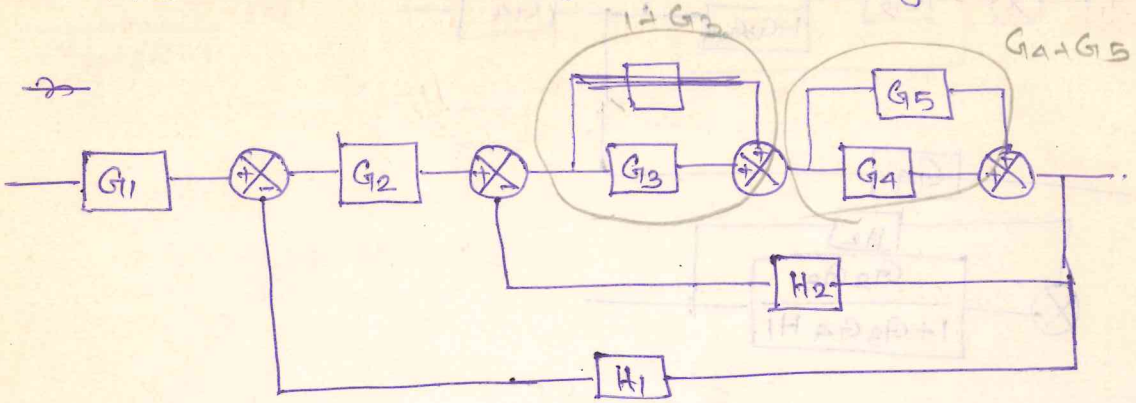
$$\frac{\dfrac{G_2G_3}{1+G_3G_4H_1}}{1+\dfrac{G_2G_3}{G_3G_4H_1}\cdot H_2} \Rightarrow \frac{G_2G_3}{1+G_3G_4H_1+G_2G_3H_2}$$

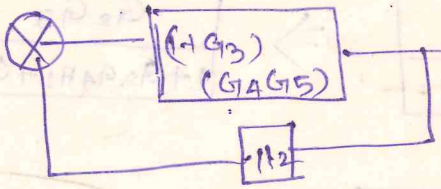


$$\frac{\dfrac{G_1G_2G_3}{1+G_3G_4H_1+G_2G_3H_2}}{1+\dfrac{G_1G_2G_3}{1+G_3G_4H_1+G_2G_3H_2}\cdot G_4} \Rightarrow \frac{G_1G_2G_3}{1+G_3G_4H_1+G_2G_3H_2+G_1G_2G_3G_4}$$

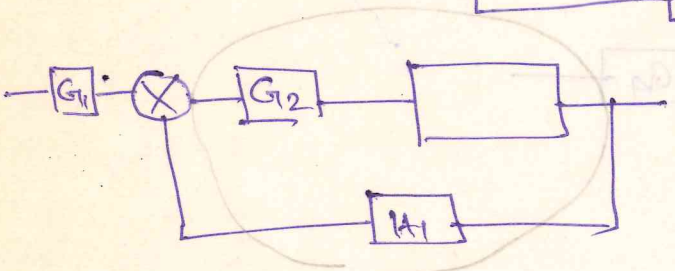
$$\frac{C(s)}{R(s)} = \frac{G_1G_2G_3G_4}{1+G_3G_4H_1+G_2G_3H_2+G_1G_2G_3G_4}$$

3) Determine the T.F for the given block diagram.





$$\Rightarrow \frac{(1+G_3)(G_4G_5)}{1+(1+G_3)(G_4G_5)H_2}$$



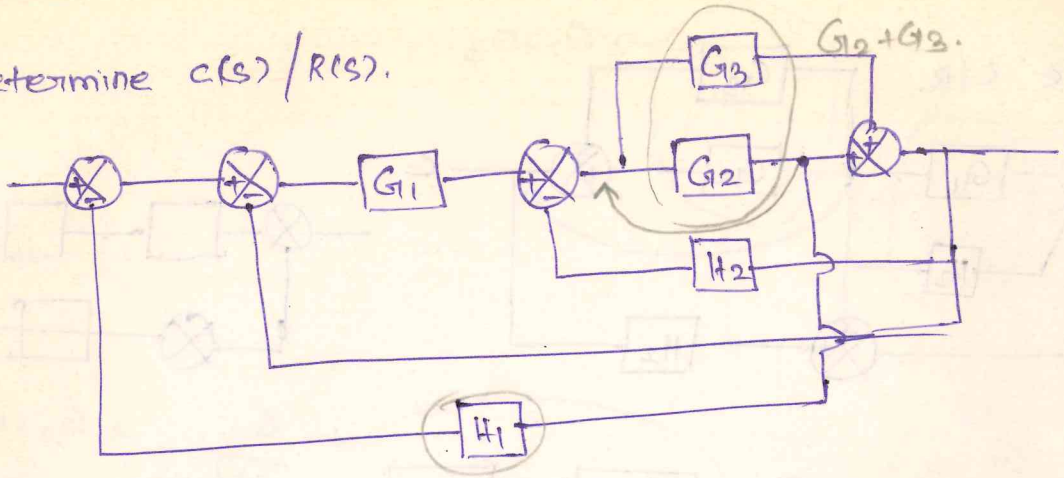
$$\frac{G_2(1+G_3)(G_4G_5)}{1+(1+G_3)(G_4G_5)H_2}$$

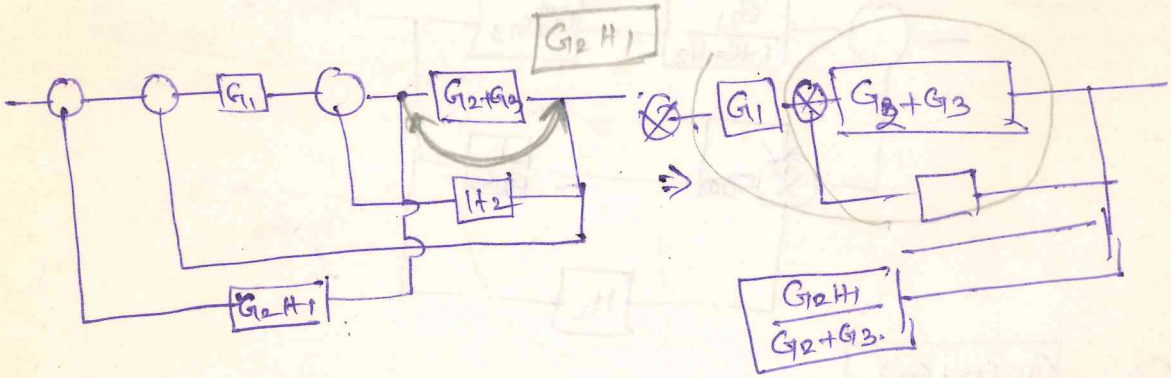
$$1+\frac{G_2(1+G_3)(G_4G_5)}{1+(G_3+1)(G_4G_5)H_2}\cdot H_1.$$

$$\Rightarrow \frac{G_2(1+G_3)(G_4G_5)}{1+(1+G_3)(G_4G_5)H_2 + G_2(1+G_3)(G_4G_5)H_1.}$$

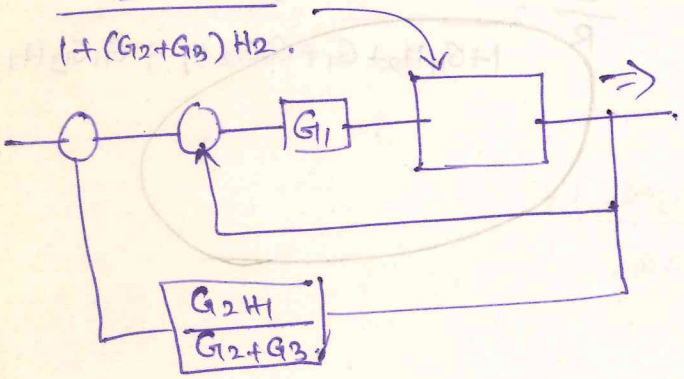
$$\frac{C(s)}{R(s)} = \frac{G_1G_2(1+G_3)(G_4G_5)}{1+(1+G_3)(G_4G_5)H_2+(1+G_3)(G_4G_5)G_2H_1.}$$

4) Determine C(s) / R(s).





$$\frac{G_2+G_3}{1+(G_2+G_3)H_2}$$

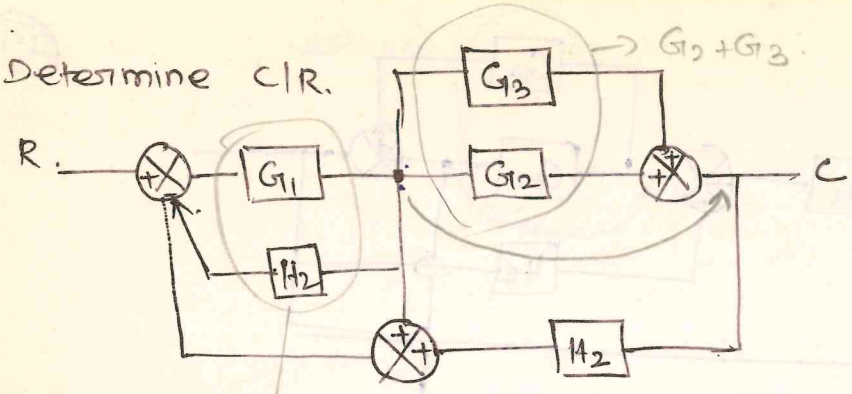


$$\frac{\dfrac{G_1(G_2+G_3)}{1+G_1(G_2+G_3)H_2}}{1+\dfrac{G_1(G_2+G_3)}{1+G_1(G_2+G_3)H_2}}$$

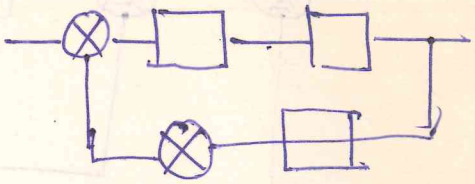
$$\Rightarrow \frac{G_1(G_2+G_3)}{1+G_1(G_2+G_3)H_2+G_1(G_2+G_3)}$$

$$\frac{C(s)}{R(s)} = \frac{G_1(G_2+G_3)}{1+G_1G_2H_1+G_1G_2+G_1G_3+G_2H_2+G_3H_2.}$$

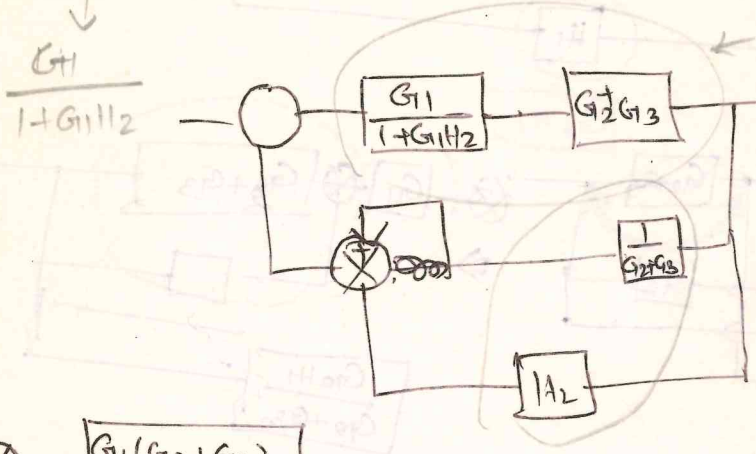
5) Determine C/R.



$G_2 + G_3$



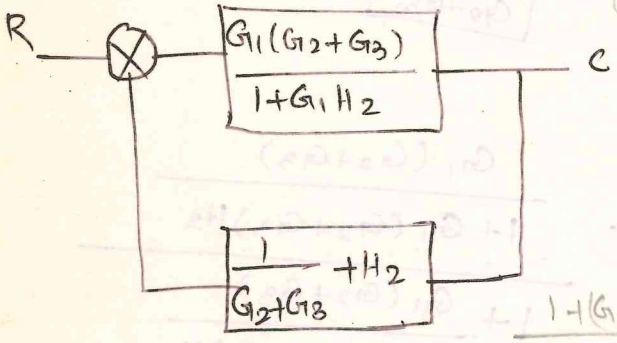
$\frac{G_1}{1+G_1H_2}$



$\frac{G_1}{1+G_1H_2} + G_2 + G_3$

$\frac{G_1(G_2+G_3)}{1+G_1H_2}$

$\frac{1}{G_2+G_3} + H_2$

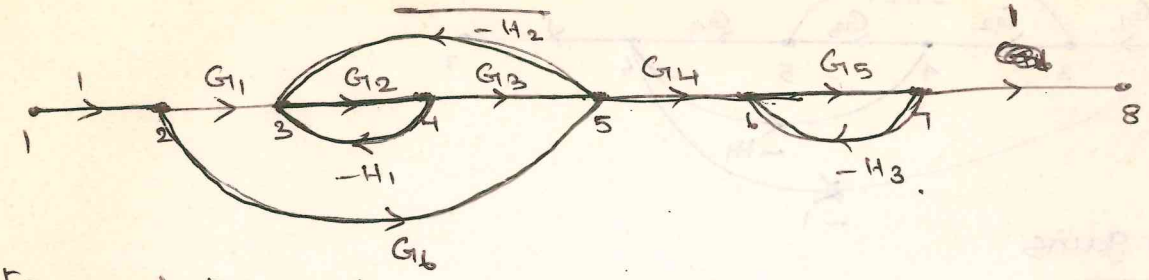


$\frac{1+(G_2+G_3)H_2}{G_2+G_3}$

$$\frac{C}{R} = \frac{G_1(G_2+G_3)}{1+G_1H_2+G_1+G_1G_2H_1+G_1G_3H_1}$$

SFG

problems.



Forward path gain

$P_1 = G_1 G_2 G_3 G_4 G_5$

$P_2 = G_4 G_5 G_6$.



Individual loop gain

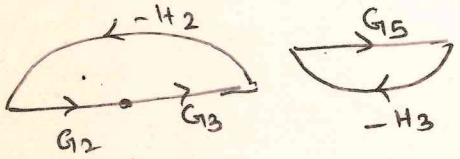


$P_{11} = -G_2 H_1$ $\quad P_{21} = -G_2 G_3 H_2$ $\quad P_{31} = -G_5 H_3$

Gain product of two non-touching loops.



$P_{12} = G_2 G_5 H_1 H_3$



$P_{22} = G_2 G_3 G_5 H_2 H_3$.

$\Delta = 1 - (P_{11} + P_{21} + P_{31}) + (P_{12} + P_{22})$

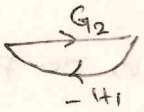
$= 1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 +$

$G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3$.

$\Delta_1 = 1$, there is no part of graph is not touching the forward path 1

$\Delta_2 = 1 - P_{11} = 1 - (-G_2 H_1)$

$= 1 + G_2 H_1$.



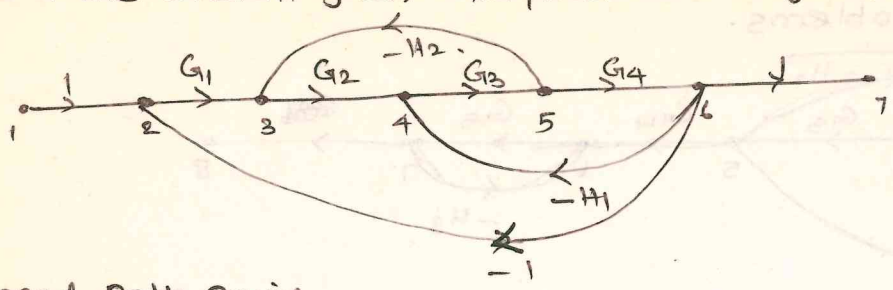
By using mason gain formula.

$T = \frac{1}{\Delta} \sum P_k \Delta_k = \frac{1}{\Delta} (P_1 \Delta_1 + P_2 \Delta_2)$.

$= \frac{G_1 G_2 G_3 G_4 G_5 (1) + G_4 G_5 G_6 (1 + G_2 H_1)}{\Delta}$.

$T = \frac{G_2 G_4 G_5 [G_1 G_3 + G_6 / G_2 + G_6 * H_1]}{\Delta}$.

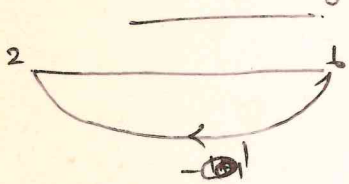
2) Find the overall gain C(s)/R(s) for the given signal flow graph



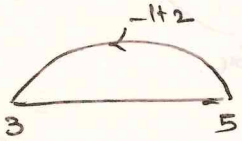
Forward path gains

$= P_1 = G_1 G_2 G_3 G_4$.

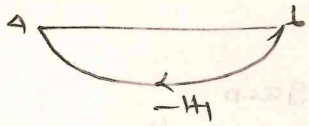
Individual loop gains.



$P_{11} = -G_1 G_2 G_3 G_4$



$P_{21} = -G_2 G_3 H_2$



$P_{31} = -G_3 G_4 H_1$

Gain product of two non touching loops

—NiL—

$\Delta = 1 - (P_{11} + P_{21} + P_{31})$

$= 1 + G_1 G_2 G_3 G_4 + G_2 G_3 H_2 + G_3 G_4 H_1$.

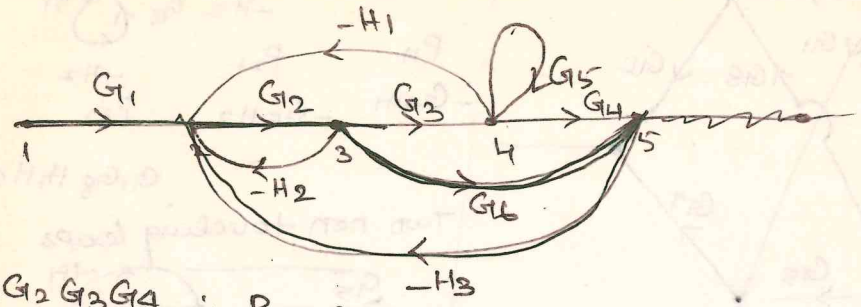
∴ Since there is no part of graph is not touching with forward path 1 so $\Delta_1 = 1$.

Mason's gain formula

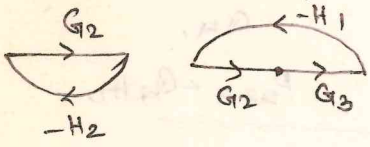
$$T = \frac{1}{\Delta} P_1 \Delta_1$$

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_1 G_2 G_3 G_4 + G_2 G_3 H_2 + G_3 G_4 H_1}$$

3) obtain C(s)/R(s)

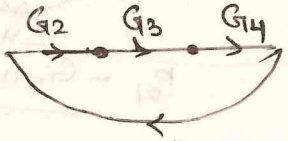


$P_1 = G_1G_2G_3G_4$; $P_2 = G_1G_2G_6$.



$P_{11} = -G_2H_2$

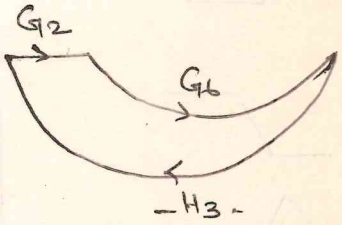
$P_{21} = -G_2G_3H_1$



$P_{31} = G_2G_3G_4H_3$

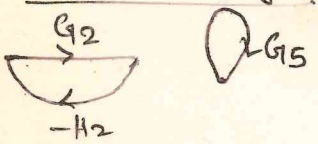


$P_{41} = G_5$

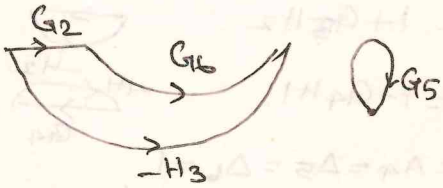


$P_{51} = -G_2G_6H_3$.

Two non touching loops.



$P_{21} = -G_2G_5H_2$

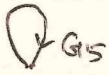


$P_{22} = -G_2G_5G_6H_3$.

$$\Delta = 1 - (P_{11} + P_{21} + P_{31} + P_{41} + P_{51}) + (P_{21} + P_{22}),$$

$$= 1 + G_2H_2 + G_2G_3H_1 + G_2G_3G_4H_3 - G_5 + G_2G_6H_3 - G_2G_5H_2 - G_2G_5G_6H_3$$

$\Delta_1 = 1$

$\Delta_2 = 1 - (G_5)$ 

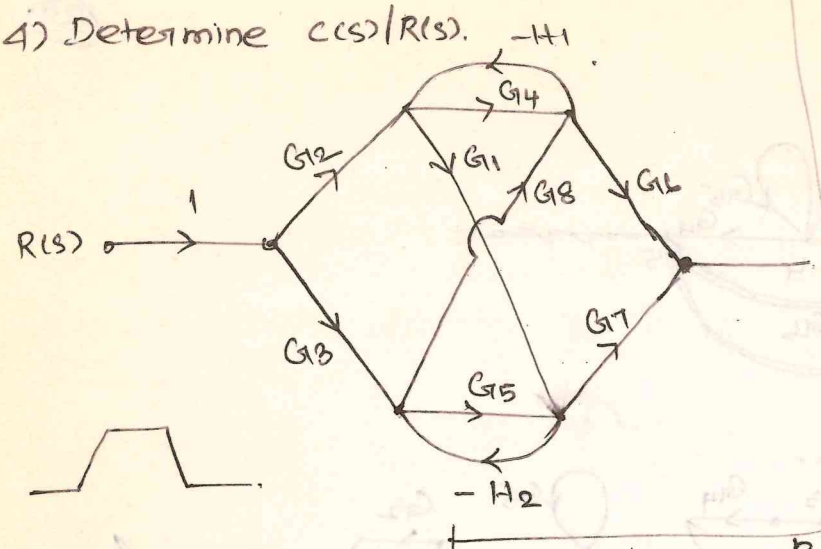
$= (1 - G_5)$

Mason gain formula

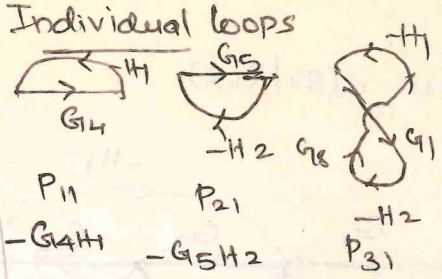
$$T = \frac{C(s)}{R(s)} = \frac{G_1G_2G_3G_4 + G_1G_2G_6(1 - G_5)}{\Delta}$$

$$= \frac{G_1G_2G_3G_4 + G_1G_2G_6 - G_1G_2G_5G_6}{\Delta}$$

4) Determine C(s)/R(s).

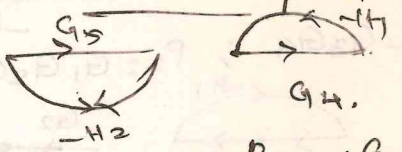


Individual loops



$P_{11} = -G_4H_1$ $\quad P_{21} = -G_5H_2$ $\quad P_{31} = G_1G_8H_1H_2$

Two non touching loops

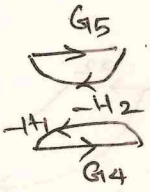


$P_{21} = -G_5H_2$ $\quad P_{22} = -G_4H_1$

$G_4G_5H_1H_2$

$\Delta = 1 + G_4H_1 + G_5H_2 + G_1G_8H_1H_2 + G_4G_5H_1H_2$

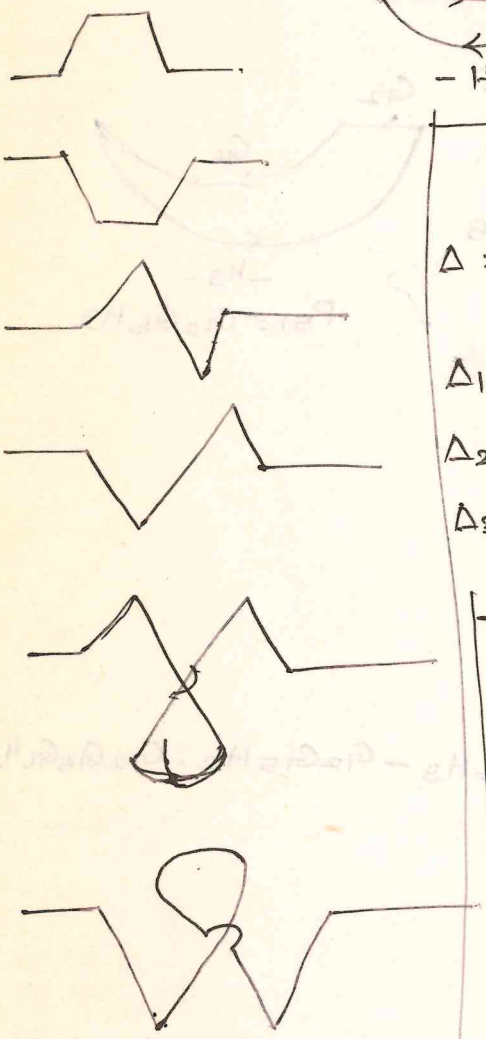
$\Delta_1 = 1 + G_5H_2$



$\Delta_2 = 1 + G_4H_1.$

$\Delta_3 = \Delta_4 = \Delta_5 = \Delta_6 = 1$

$$T.F = \frac{G_1G_4G_6(1+G_5H_2) + G_3G_5G_7(1+G_4H_1) + G_1G_2G_7 + G_3G_6G_8 + G_1G_3G_7G_8H_1 - G_1G_2G_6G_8H_2}{1 - G_4H_1 + G_5H_2 + G_1G_8H_1H_2 + G_4G_5H_1H_2.}$$



$P_1 = G_1G_4G_6$

$P_2 = G_3G_5G_7$

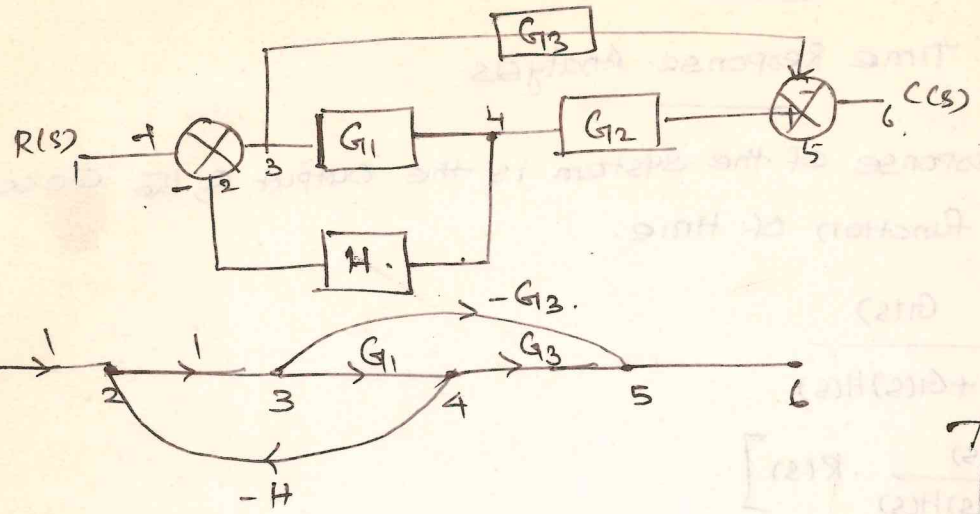
$P_3 = G_1G_2G_7$

$P_4 = G_3G_6G_8$

$P_5 = -G_1G_3G_7G_8H_1$

$P_6 = -G_1G_2G_6G_8H_2.$

5) Determine C(s)/R(s).





$$\frac{G_1G_2 + G_3}{1 + G_1H}$$

$P_1 = G_1G_2$

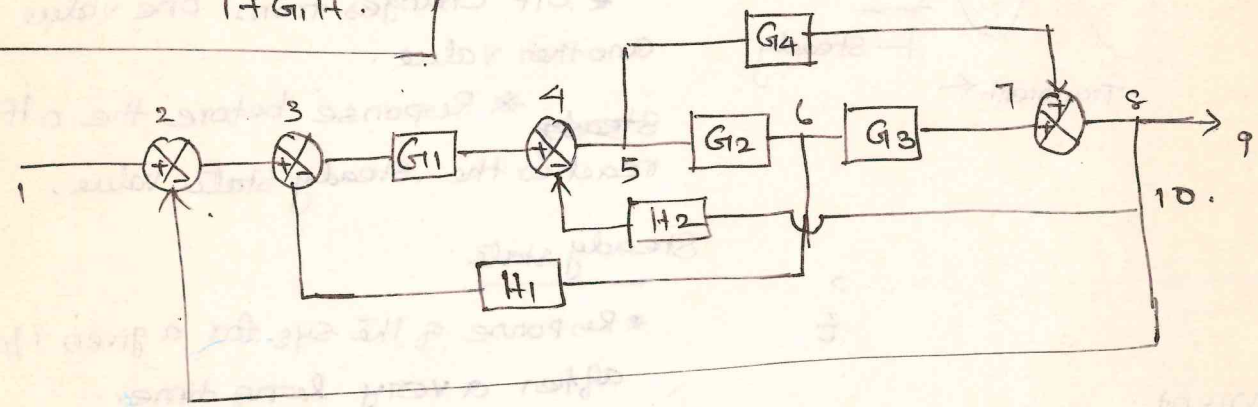
$P_2 = -G_3$.

One individual loop $P_{11} = -G_1H$

$\Delta = 1 + G_1H$.

$$T.F = \frac{G_1G_2 - G_3}{1 + G_1H}$$

6)



No. of forward paths : 2

Individual loops : 5.

$$T = \frac{G_1G_2G_3 + G_1G_4}{1 + G_1G_2G_3 + G_1G_2H_1 + G_2G_3H_2 + G_1G_4 + G_4H_2}$$